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Plan curvature and landslide probability in regions dominated by earth flows and earth slides

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Abstract

Damaging landslides in the Appalachian Plateau and scattered regions within the Midcontinent of North America highlight the need for landslide-hazard mapping and a better understanding of the geomorphic development of landslide terrains. The Plateau and Midcontinent have the necessary ingredients for landslides including sufficient relief, steep slope gradients, Pennsylvanian and Permian cyclothems that weather into fine-grained soils containing considerable clay, and adequate precipitation. One commonly used parameter in landslide-hazard analysis that is in need of further investigation is plan curvature. Plan curvature is the curvature of the hillside in a horizontal plane or the curvature of the contours on a topographic map. Hillsides can be subdivided into regions of concave outward plan curvature called hollows, convex outward plan curvature called noses, and straight contours called planar regions. Statistical analysis of plan-curvature and landslide datasets indicate that hillsides with planar plan curvature have the highest probability for landslides in regions dominated by earth flows and earth slides in clayey soils (CH and CL). The probability of landslides decreases as the hillsides become more concave or convex. Hollows have a slightly higher probability for landslides than noses. In hollows landslide material converges into the narrow region at the base of the slope. The convergence combined with the cohesive nature of fine-grained soils creates a buttressing effect that slows soil movement and increases the stability of the hillside within the hollow. Statistical approaches that attempt to determine landslide hazard need to account for the complex relationship between plan curvature, type of landslide, and landslide susceptibility.

Keywords: Engineering geology; Geologic hazards; Geomorphology; Landslides; Statistical analysis; Mechanical analysis

1. Introduction

The Appalachian Plateau in Pennsylvania, West Virginia, and Ohio, USA (Fig. 1) is an area of high landslide incidence in the eastern United States (Radbruch-Hall et al., 1982). Landslides in this region have destroyed houses, closed highways, and damaged infrastructure. In March of 1975, a landslide in the town of McMechen, West Virginia damaged 56 houses on a colluvial hillside (Gray and Gardner, 1977). Geologically, this portion of the Appalachian Plateau features Pennsylvanian and Permian cyclothems. Cyclothems are rhythmic sedimentary sequences consisting of alternating transgressive and regressive deposits (Heckel, 1977). In southeastern Ohio, the dip of the units is about 6 m/km (~0.3°) to the southeast (Fisher et al., 1968). These cyclic sedimentary deposits contain paleosols and shales that weather to form clayey residual soils that are prone to mass movement. This region is incised by the

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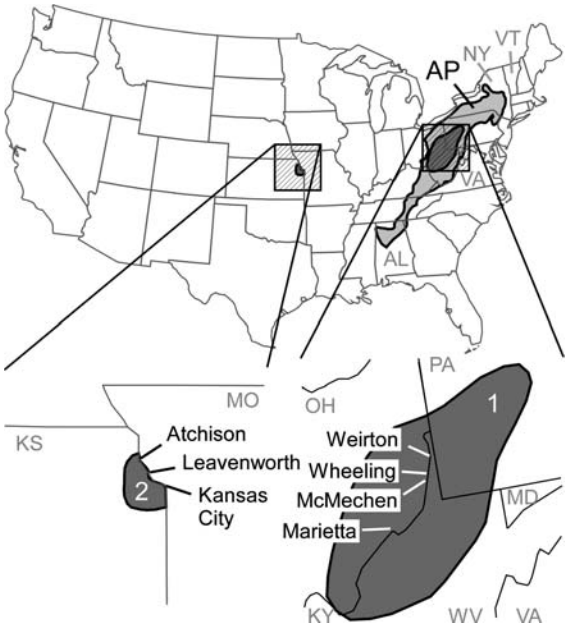

Fig. 1. Location map. Study area 1 is a region with a high incidence of landslides within the Appalachian Plateau Province (AP) and includes parts of Ohio (OH), West Virginia (WV), and Pennsylvania (PA). Study area 2 is located in northeastern Kansas (KS). Other states shown include Alabama (AL), Missouri (MO), Maryland (MD), New York (NY), Vermont (VT), Virginia (VA), and Kentucky (KY).

Ohio River and its tributaries creating the necessary slope gradients and relief, and receives sufficient precipitation for almost annual landslide activity.

Other areas in the central United States have conditions analogous to the Appalachian Plateau and have varying degrees of landslide susceptibility. One analogous region is the Kansas City Metropolitan Area in Missouri and Kansas (Fig. 1) where nearly horizontal Pennsylvanian cyclothems are incised by the Missouri and Kansas rivers. The dip of the units is about 3 m/km (~0.2°) to the west-northwest (Ohlmacher, 2000b). The mean annual precipitation for Kansas City (~97 cm/year) is about the same as the Appalachian Plateau (97–112 cm/year) (National Climatic Data Center, On-line Data). The local relief around Kansas City is less than that found in the Appalachian Plateau, but this has not prevented the occurrence of major landslides. In 1995, two new houses were destroyed by a landslide in Overland Park, Kansas, a suburb of Kansas City.

The development of quantitative landslide-susceptibility and landslide-hazard maps is a topic of active research (Soeters and Van Westen, 1996). Many different approaches are being used, including multiple logistic regression (Ohlmacher and Davis, 2003), artificial neural networks (Lee et al., 2003, 2004; Yesilnacar and Topal, 2005), factor weighting (Lee and Min, 2001; Süzen and Doyuran, 2004a,b), decision trees (Keefer, 1993), fuzzy set theory (Chung and Fabbri, 2001), Bayesian statistics (Chung and Fabbri, 1998; Lee and Choi, 2004), and other approaches. The discussion of the advantages and disadvantages of the various methods is not germane to this paper. Here, the primary focus is the datasets used in the analysis. Some researchers use every dataset available regardless of how it relates to landslide susceptibility. These individuals collect data from a variety of sources, analyze the data with a statistical method, and get a set of values that measure the relative susceptibility. This approach may give accurate results for conditions that fall within the bounds of the data; however, it does not provide further insight into landslide processes.

Some datasets contain variables that change over time. For example, land use may be highly correlated with landslide susceptibility, but land use patterns change in space and time. If land use is a parameter in landslide susceptibility analysis, a commitment is being made to reevaluate the landslide susceptibility as land use changes. Additionally, the initial landslide susceptibility map may encourage some people to alter design and construction practices in areas where the hazard is higher. Newly developed areas may have a different susceptibility than older developed areas with otherwise similar characteristics. In such a case it would not be sufficient to simply substitute a new land-use map; instead, the entire statistical analysis will need to be repeated.

In Kansas, the choice of parameters used in the statistical analysis has been guided by engineering mechanics. Slope failure is controlled by landslide geometry, material properties, and pore-water pressure. Pore-water pressure, in general, varies considerably over short distances and short increments of time, and the change in pore pressure in response to rainfall is related to the intensity and duration of the rainfall, the time since the last rainfall, and the pre-storm moisture in the soil (Haneberg and Gökce, 1994). Because pore-water pressure is highly variable, it is ignored in the landslide susceptibility analysis used in Kansas. Slope geometry and material properties are less variable. A recent landslide hazard analysis in Kansas used the slope gradient and the surface geology as the parameters for failure-surface geometry and material properties, respectively (Ohlmacher and Davis, 2003). This approach has produced usable landslide-hazard maps for the Kansas portion of the Kansas City metropolitan area.

Another factor that can be considered in landslide-hazard analysis is topographic curvature. Mathematically, curvature is defined as the change in slope angle along a

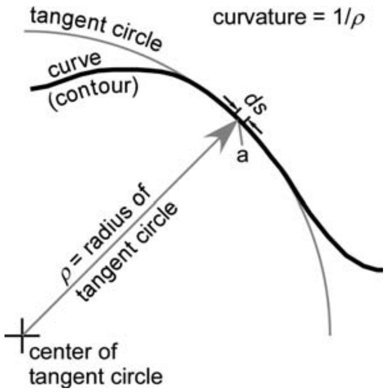

Fig. 2. Mathematical definition of curvature. Curvature is the change in slope of a curve over a small increment *ds* along the curve at point a. The curvature is the inverse of the radius of a circle ρ that is tangent to the curve over the same increment *ds*.

very small arc of the curve, *ds*, (Fig. 2) (Thomas, 1968). Curvature is the inverse of the radius of a circle that is tangent over a small arc, at least three points, of the curve (Kepr, 1969). A straight line has a curvature of zero (infinite radius of the tangent circle) and as the curve becomes more acute the curvature value increases. At a point on a three-dimensional surface — for example, a hillslope, an infinite number of curvature values can exist, because the curvature is a function of the orientation of the increment *ds* and the orientation of the plane of intersection with the surface. A special case exists where the hillslope at a point is straight in all directions and the curvature is equal to zero in all directions. The three curvature values used for hillslope and landslide analysis are profile, plan, and tangential curvature (Dikau, 1989; Moore et al., 1993a,b; Ayalew and Yamagishi, 2004). Profile curvature is the curvature in the downslope direction (aspect) along a line formed by the intersection of an imaginary vertical plane with the ground surface. For profile curvature the increment *ds* has the same bearing and dip angle as the maximum slope gradient at the point of interest. Plan curvature is the curvature of the topographic contours or the curvature of a line formed by the intersection of an imaginary horizontal plane with the ground surface. Tangential curvature is the curvature in a vertical plane that is tangent to the contour at the point of interest. The orientation of the increment *ds* is horizontal and normal to the aspect for both the plan and tangential curvatures.

The curvature of a line formed by the intersection of a surface and an imaginary plane is given by:

$$\kappa = \frac{f_{xx}\cos^2\beta + 2f_{xy}\cos\beta\sin\beta + f_{yy}\sin^2\beta}{(\sqrt{q})\cos\varphi} \quad (1)$$

where: $q=f_x^2+f_y^2+1$, $f_x=\partial f/\partial x$, $f_y=\partial f/\partial y$, $f_{xx}=\partial^2 f/\partial x^2$, $f_{yy}=\partial^2 f/\partial y^2$, $f_{xy}=\partial^2 f/\partial x\partial y$, β is the angle between the tangent of the curve and the x-axis in the horizontal plane and, φ is the angle between the normal to the surface and the section plane (the imaginary plane of intersection) (Kepr, 1969; Mitášová and Hofierka, 1993; Moore et al., 1993a). For profile curvature, κ_{profile}, the normal to the surface is always in the section plane; the angular relationships are

$$\cos\varphi = 1, \sin\beta = \frac{f_x}{\sqrt{pq}}, \text{ and } \sin\beta = \frac{f_y}{\sqrt{pq}}; \text{ and}$$

Eq. (1) becomes

$$\kappa_{\text{profile}} = \frac{f_{xx}f_x^2 + 2f_{xy}f_xf_y + f_{yy}f_y^2}{p\sqrt{q^3}}$$

where $p=f_x^2+f_y^2$ (Mitášová and Hofierka, 1993; Moore et al., 1993a). For plan curvature, κ_{plan}, the normal to the curve is only in the section plane when the ground surface is vertical. The angular relationships are

$$\cos\varphi = \sqrt{\frac{p}{q}}, \cos\beta = \frac{f_y}{\sqrt{p}}, \text{ and } \sin\beta = -\frac{f_x}{\sqrt{p}}; \text{ and}$$

the equation is

$$\kappa_{\text{plan}} = \frac{f_{xx}f_y^2 - 2f_{xy}f_xf_y + f_{yy}f_x^2}{\sqrt{p^3}}$$

(Mitášová and Hofierka, 1993; Moore et al., 1993a). For tangential curvature, $\kappa_{\text{tangential}}$, the normal to the curve is only in the section plane when the ground surface is horizontal. Thus, the angular relationships are

$$\cos\varphi = \frac{1}{\sqrt{q}}, \cos\beta = \frac{f_y}{\sqrt{p}}, \text{ and } \sin\beta = -\frac{f_x}{\sqrt{p}}; \text{ and}$$

the equation is

$$\kappa_{\text{tangential}} = \frac{f_{xx}f_y^2 - 2f_{xy}f_xf_y + f_{yy}f_x^2}{p}.$$

This corrects the error in the angle φ found in published tangential curvature equations by Mitášová and Hofierka (1993) and Gallant and Wilson (1996).

These curvature equations were tested on mathematical surfaces created from a series of functions $Z_{(x,y)}$. A 3×3 grid was created with the point of interest at the center of the grid, and the required derivatives of $Z_{(x,y)}$

were calculated using published approximations (Mitášová and Hofierka, 1993; Moore et al., 1993a; Gallant and Wilson, 1996). Additionally, the curvature was calculated from the derivatives of function $Z_{(x,y)}$ evaluated at the center point of the grid, and graphically by determining the radius of the tangent curve. The results from the three approaches compare favorable indicating that the curvature equations presented above produce reasonable results. As the spacing between grid points decreased, the accuracy of the approximations of the derivatives improves along with the values for the curvature. One function $Z_{(x,y)}$ was for a series of concentric circular contours, and the plan curvature equations using the approximations of the derivatives correctly calculated the inverse of the radius of the circular contours.

Several authors (Dietrich et al., 1995; Heimsath et al., 1999, 2005) use the del^2 operator ($\nabla^2 z = \partial^2 z/\partial x^2 + \partial^2 z/\partial y^2$) as a curvature value. By examining Eq. (1), the del^2 operator can be a curvature value if the following conditions are met: $\partial^2 z/\partial x \partial y = 0$, $\beta = 45°$, and $\cos\varphi = 2/\sqrt{q}$. If the term $\partial^2 z/\partial x \partial y$ is not equal to zero and the del^2 operator is used, then the magnitude and possibly the sign of the curvature value will be incorrect. The sign of the curvature value is important for determining concavity or convexity of the curve. Points exist within a digital elevation model where $\partial^2 z/\partial x \partial y$ is not equal to zero and where use of the del^2 operator will give erroneous results.

Curvature equations also have been developed by Zevenbergen and Thorne (1987). The plan curvature equation published in Zevenbergen and Thorne is calculating tangential curvature. The profile-curvature equation is missing the $\sqrt{q^3}$ term in the denominator, and thus the values are incorrect.

Both profile and plan curvature will affect the susceptibility to landslides. Profile curvature affects the driving and resisting stresses within a landslide in the direction of motion. Plan curvature controls the convergence or divergence of landslide material and water in the direction of landslide motion (Carson and Kirkby, 1972). This paper examines the role of plan curvature in landslide susceptibility and the effect of convergence of material in the direction of landslide motion.

2. Landslide background

This paper uses the landslide classification scheme of Varnes (1978) and Cruden and Varnes (1996) that subdivides landslides into types based on material and movement. The study areas are covered predominantly by fine-grained soils referred to as earth, and the predominant landslide types are earth slides (including slumps), earth flows, and combination earth slide-flow (Sharpe and Dosch, 1942; Fisher et al., 1968; Mills et al., 1987; Ohlmacher, 2000b). Other types of landslides, especially rock falls and rock topples, occur in these regions but are not considered in this analysis.

The Appalachian Plateau (Fig. 1) is a dissected plateau that stretches from central New York State to northern Alabama in the eastern United States. The region of particular interest to this study is the portion with a high-incidence of landslides labeled 1 in Fig. 1 (Radbruch-Hall et al., 1982). Jacobson and Pomeroy provide an overview of landslides in the Appalachian Plateau (Mills et al., 1987). Along the Ohio River near Weirton, West Virginia, the local relief is about 128 m (D'Appolonia et al., 1967). The U.S. Geological Survey Wheeling 7½-minute topographic quadrangle has a total relief of 231 m and the local relief along the Ohio River is up to 212 m. Further south, the Marietta, Ohio 7½-minute topographic quadrangle has a total relief of 135 m and bluffs along the Ohio and Muskingum rivers can have local relief values of up to 85 m. The Marietta and Wheeling areas were chosen as examples of the Appalachian Plateau, because of landslide-inventory mapping by Pomeroy (1984) and Lessing et al. (1976), respectively.

The Midcontinent region in the United States is an area of generally low relief that extends from the Appalachian Plateau to the Rocky Mountains and which is locally dissected by rivers and streams. Ohlmacher (2000b) gives a description of landslides in the vicinity of the City of Atchison in northeastern Kansas (Fig. 1). As an example of the Midcontinent region, the Leavenworth, Kansas 7½-minute topographic quadrangle has a total relief of 114 m and the bluffs along the Missouri River and its tributaries can have local relief values up to 85 m. It should be noted that the Oread Escarpment, a cuesta where the southeast-facing slopes are steeper and have higher relief than the northwest-facing slopes, intersects the Missouri River valley at the City of Leavenworth creating locally higher relief. Away from the escarpment, the bluffs of the Missouri River have modest relief values of about 45 m.

The bedrock of both areas consists of cyclically deposited Pennsylvanian and Permian sedimentary rocks called cyclothems (Heckel, 1977). An ideal cyclothem contains a transgressive and regressive sequence with sandstones, siltstones, shales, and limestones (Fig. 3). The ideal sequence begins with such non-marine sedimentary rocks as sandstones, siltstones, gray and red shales, and coal. The sequence proceeds into marine limestones and shales with black shales representing the end of the transgression. The regression is represented

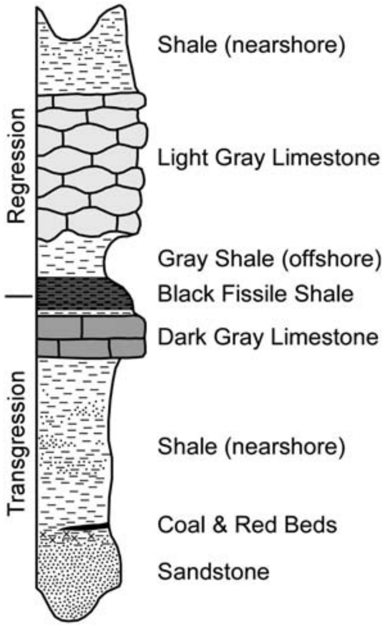

Fig. 3. Idealized cyclothem simplified from (Heckel, 1977).

by marine limestones and shale grading into non-marine sedimentary rocks. Although the basic sequence of layers is relatively constant, cyclothems can vary both spatially and temporally; for example, the non-marine sequence can be thicker in some regions or during some time intervals or cyclothems can be incomplete missing one or more layers. Of particular interest to landslide studies are the shales that can be zones of weakness within the bedrock and can weather to provide clayey colluvium (Fisher et al., 1968; Hamel and Flint, 1972). Slope stability problems of the Appalachian Plateau are related to the varied physical characteristics of the cyclothems with the non-marine shales being the weakest units (D'Appolonia et al., 1967).

Shales, disaggregated in the laboratory, and soil samples from Kansas are classified as CH, CL, and ML soils and contain expansive clays including mixed-layer illite/smectite, mixed-layer chlorite/smectite, and Ca-montmorillonite along with non-expansive illite, kaolinite, and chlorite (Ohlmacher, 2000b). The red mudstones and shales in eastern Ohio have a tendency to slake rapidly; however, other shales will slake to varying degrees (Fisher et al., 1968). Soil developed from the shale and sampled in the Appalachian Plateau region have peak cohesion values of 0–38 kPa, peak angle of internal friction of 19°–30°, residual cohesion values of 0–3 kPa, and residual angle of internal friction of 8°–25° (D'Appolonia et al., 1967; Gray and Gardner, 1977; Hamel and Adams, 1981).

3. Plan curvature and landslides

Plan curvature can be used to subdivide hillslopes into three regions: hollows, noses, and relatively planar regions (Fig. 4). Hollows are regions in which the plan curvature of the contours is concave in the downslope direction and where surface water would converge as it moves downslope (Hack and Goodlett, 1960; Reneau and Dietrich, 1987). The term "coves" was used by Sharpe and Dosch (1942) and Patton (1956) and is analogous to "hollows". Noses are regions where the plan curvature of the contours is convex in the downslope direction, and surface water will diverge (Hack and Goodlett, 1960). Relatively planar regions (Jacobson et al., 1993) have plan curvature values around zero. Hack and Goodlett (1960) used the term "side slopes" for these regions. For the purpose of this paper, no cutoff values of curvature were established between the hollows, planar regions, and noses.

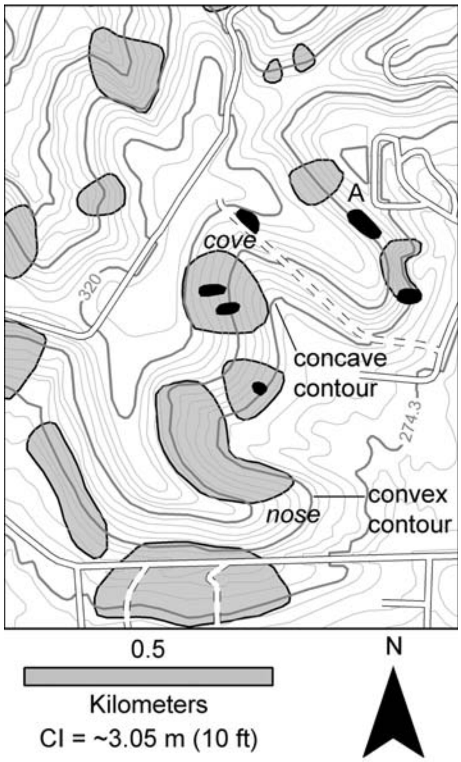

Fig. 4. Hollows, noses and landslide features. Hollows are regions with concave topographic contours; while noses are regions of convex contours. The black polygons are recent landslides having fresh landslide features (scarps and fissures). The gray polygons are older landslides having muted features. The landslides in this area are earth flows and earth slides. The area labeled A is a region of planar plan curvature with a recent and an older landslide. This landslide inventory map is located within the City of Leavenworth, Kansas (Fig. 1).

3.1. Field observations

Sharpe and Dosch (1942) observed in Ohio that earth flows tend to occur in valley-head coves (hollows). Hollows concentrate ground and surface water, and the concentration of water probably leads to increased earth-flow activity (Patton, 1956; Carson and Kirkby, 1972). Carrara et al. (1977) did a statistical analysis of landslides and morphologic features for two study areas in Calabria, Italy, and reported that in the first study area hillslopes with planar plan curvature had a higher percentage of landslides than hollows, 55% to 34% respectively. In the second study area hollows had a higher percentage of landslides than hillsides with planar plan curvature, 37% to 53% respectively. In both study areas noses had the lowest percentage of landslides. In Appalachian Plateau of West Virginia, the frequency of landslides is greatest in concave landforms with almost twice as many landslides in concave areas as in planar regions (Lessing et al., 1976; Lessing and Erwin, 1977; Lessing et al., 1994). For the Greater Pittsburgh, Pennsylvania region, Pomeroy (1982) stated that 60% of the landslides occur in concave landforms with 25% occurring on planar slopes. None of the papers that are based on work in the Appalachian Plateau describes how curvature was measured. East of the Appalachian Plateau (Fig. 1) in the Valley and Ridge Province of West Virginia and Virginia, Jacobson et al. (1993) observed that hillsides with planar plan curvature have the highest susceptibility to landslides. Their study area included debris slides, earth slides, debris flows, and earth flows.

Field observations reveal that earth flows can occur in any area of the slope regardless of plan curvature (Fig. 4). As observed in earlier studies, earth flows do occur in hollows-for example, in Kansas (Fig. 4) and Ohio (Fig. 5a). Examples are presented of recent earth flows in regions of plan curvature that are broadly concave (Fig. 5b) and broadly convex (Fig. 5c) in eastern Ohio. Label A in Fig. 4 indicates a region of planar plan curvature with both recent and older earth flows. Numerous hollows exist without any recent or older landslide features (Fig. 5d). What is the distribution of earth flows with respect to plan curvature? Are hollows an end-member landform in a terrain dominated by earth flows and earth slides?

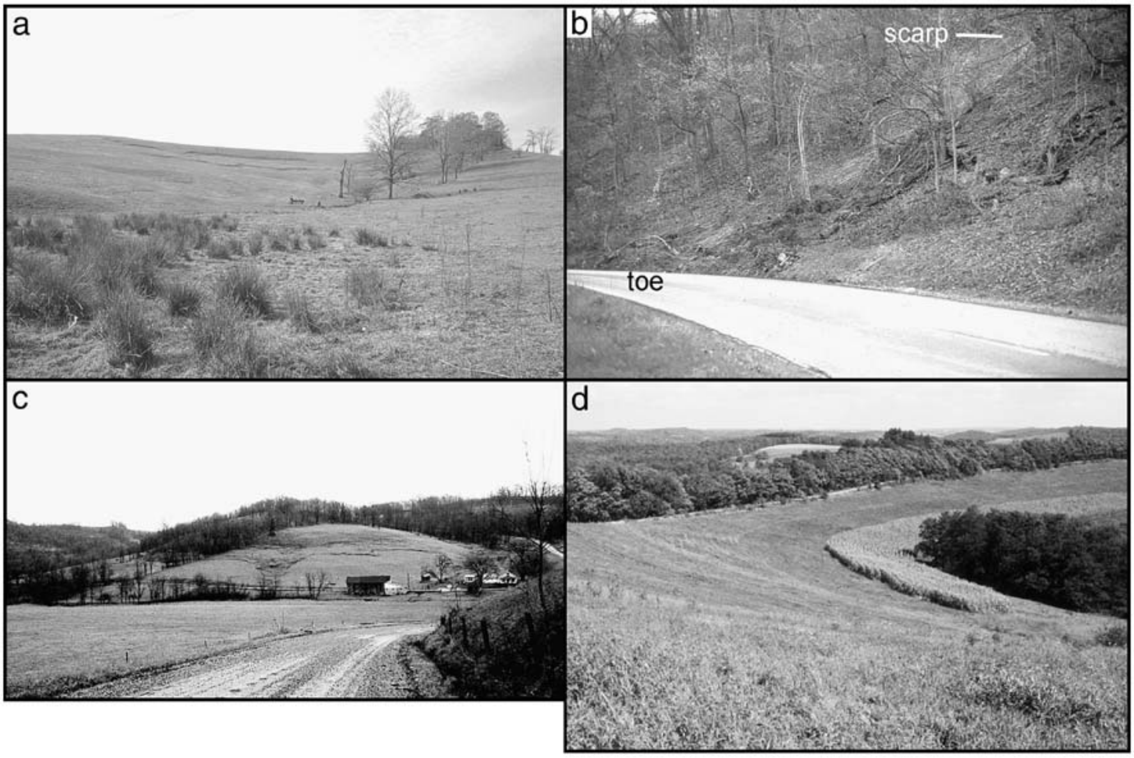

Fig. 5. Photographs of landslides and plan curvature. (a) A recent earth flow that occurred in a hollow (concave plan curvature) located eastern Ohio. (b) A recent earth flow that occurred in a broadly concave (almost planar plan curvature) located across the Ohio River from Wheeling, West Virginia. (c) A recent earth flow that occurred on nose (convex plan curvature) located in eastern Ohio. (d) A hollow located northeast of Wheeling, West Virginia with no evidence of recent or older landslide activity.

3.2. Statistics background

Plan curvature was calculated from U. S. Geological Survey Digital Elevation Model (DEM) data using a computer program based on the curvature equations presented above. The program input is a DEM that was converted to an ESRI raster in ASCII format. The ESRI raster is subdivided into 3 × 3 clusters of grid cells, and the required derivatives are calculated using the equations found on page 445 of Moore et al. (1993a). The derivatives are then entered into the equation for plan curvature. If one or more elevation values are missing from a 3 × 3 cluster, no curvature value is calculated for that cluster, and no curvature values are calculated for the grid cells along the perimeter of the database. The output is an ESRI raster in ASCII format with plan curvature values. Positive values of curvature are concave in a downslope direction or hollows. A comparison of calculated plan curvature values calculated using a 30-m DEM and topographic contours is presented for a hollow in Fig. 6.

Both 10-m and 30-m U. S. Geological Survey DEM data were used in the analysis. In Kansas, 10-m DEM datasets are only available for a portion of the study area. A comparison of the 30-m and 10-m DEM datasets was conducted for a 64-km^2 region consisting of the Kansas portion of the U. S. Geological Survey Parkville and North Kansas City 7½-minute topographic quadrangles. As would be expected, the values for the grid cells vary between the 10-m and 30-m data because derivatives calculated from the 10-m data are based on more closely spaced data than those calculated from 30-meter data. The range of curvature values is larger for the 10-m DEM dataset with curvature values from −8.8 to 9.2 in units of 1/m whereas the values for the 30-m dataset range from −1.5 to 2.7. The relative distribution of the plan curvature values is roughly the same for both data sets (Fig. 7a). Only a few data points exist near the positive (>0.15) and negative (<−0.15) extremes of both datasets in Fig. 7a. Thirty-meter DEM datasets were used only where 10-m DEM datasets were unavailable.

Plan curvature is a continuous variable as opposed to a categorical variable, for example, geologic unit. In order to evaluate the probability of a landslide given the plan curvature, the plan curvature values were

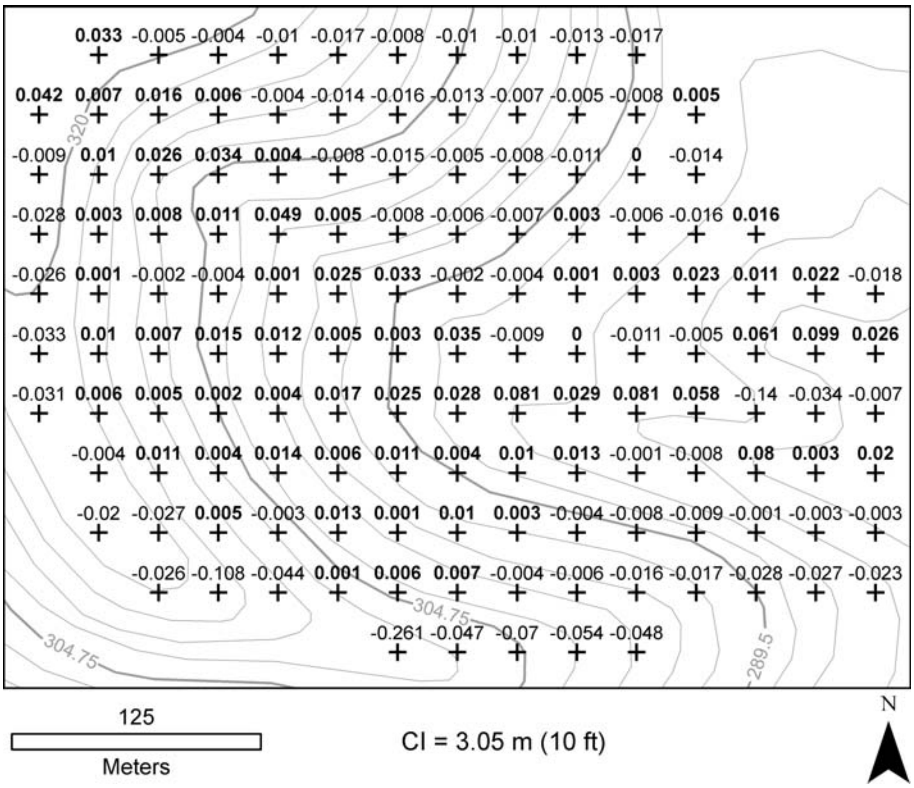

Fig. 6. Plan curvature values. This map of a hollow near Leavenworth, Kansas (Fig. 1) shows plan curvature values calculated using the program written for this study. Positive plan curvature values are concave regions or hollows. Plan curvature values greater than or equal to zero are shown in bold font. The DEM has 30-m spacing between elevation values.

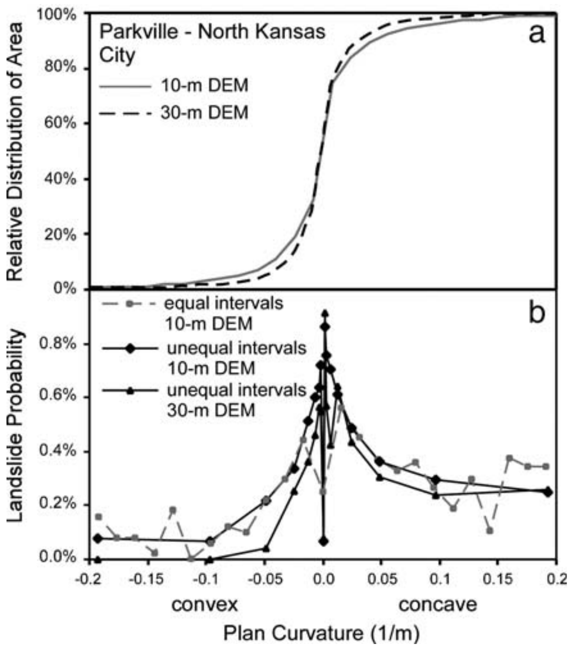

Fig. 7. Comparison of 10-m and 30-m DEM data. Graph (a) shows the relative distribution of area versus plan curvature in a cumulative form for the both DEM datasets. Graph (b) shows a comparison of landslide probability between equal and unequal intervals of plan curvature values and between the 10-m and 30-m DEM data. The DEM data is for the U. S. Geological Survey Parkville and North Kansas City, Kansas and Missouri 7½-minute topographic quadrangles.

subdivided into intervals. The probability was then calculated using Bayes' Theorem:

$$P(\text{landslide}|\text{curvature}) = \frac{P(\text{curvature}|\text{landslides}) \cdot P(\text{landslides})}{P(\text{curvature})}$$

where $P(\mathrm{A}|\mathrm{B})$ is the conditional probability, the probability that A will occur given that B has occurred, and $P(\mathrm{A})$ is the probability that A will occur (Davis, 1973; Carr, 1995). The probabilities are calculated using the areal extent of the parameters (landslides and plan curvature intervals), where P(curvature|landslides) for a range of curvature values is found by dividing the number of grid cells with landslides within that curvature range by the total number of grid cells with landslides, P(landslides) is the total number of grid cells with landslides divided by the total number of grid cells, and P(curvature) is the number of grid cells in each curvature range divided by the total number of grid cells.

A fluctuation in the probability values is observed when intervals of plan curvature with equal ranges of values were used (Fig. 7b), because the intervals close to zero had very large numbers of grid cells and the intervals near the positive and negative extremes had a very few cells. Additionally, landslides in two map areas in Kansas comprise only 0.5% of the total area. This low percentage of landslides for the study area leads to high fluctuations in landslide probabilities in intervals with few cells (Fig. 7b). In order to increase the number of grid cells in intervals with few cells and to improve the probability estimate for each interval, the total range of values was subdivided into intervals with unequal ranges of plan curvature. The subdivision that worked best had a very small interval with a plan curvature value of zero at the center of the range and each adjacent interval being twice the range as the interval before. For these studies, the range centered on a plan curvature of zero was set to 0.001 (1/m). The ranges for intervals on either side were 0.002; the next range was 0.004, and the ranges continued to double away from the zero interval. Fig. 7b displays a comparison between the results using equal range intervals and intervals with unequal ranges of values. Between plan curvature values of −0.05 and 0.05, the equal and unequal intervals compare favorably except for the interval containing a plan curvature of zero. For the regions below −0.05 and above 0.05, the probability values for the intervals with equal ranges display considerably more scatter. This is the region where the number of cells in the interval is low (Fig. 7a). For the remaining analysis, intervals with unequal ranges as defined above will be used.

The interval of plan curvature containing zero has a landslide probability value that is considerably lower than the two adjacent plan curvature intervals (all data curve in Fig. 8). Part of the reason for this anomaly is that almost all the low slope areas including valley floors and ridge crests have plan-curvature values near zero. This fact combined with the lack of landslides in

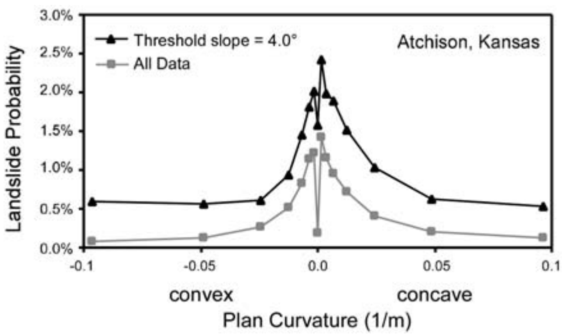

Fig. 8. Threshold slope comparison. The gray line incorporates all the data from the Atchison East and West, Kansas and Missouri 7½-minute topographic quadrangles. The black line has the grid cells with slope gradients less than 4° removed from the data.

Table 1
Landslide data

	Recent landslides			Older landslides			Map area (km^2)
	Number	Average area (m^2)	Total area (km^2)	Number	Average area (m^2)	Total area (km^2)	
Atchison East and West, KS	145	880	0.13	85	11,600	0.98	202.35
Easton East and West, KS	550	530	0.29	880	9900	8.71	299.19
Leavenworth, KS	226	880	0.20	430	16,300	7.03	171.42
McLouth and Jarbalo, KS	553	600	0.33	554	7700	4.3	299.60
Parkville and North Kansas City, KS	65	330	0.02	95	1900	0.18	63.69
Potter and Oak Mills, KS	329	750	0.25	604	11,600	7.00	279.10
Marietta, OH (Pomeroy, 1984)	708	2500	1.78	–	–	–	111.85
Wheeling, WV (Lessing et al., 1976)	290	2300	0.67	971	10,900	10.60	122.39
Wheeling, WV and OH (Davies and Ohlmacher, 1978)	78	14,100	1.10	314	122,300	38.42	147.41

regions of low slope lowers the overall probability for the interval including a plan curvature value of zero. One solution is to use a minimum slope-gradient threshold value. A dataset where grid cells with the slope gradient below a minimum threshold of 4° were removed has an increase in the probability of landslides for the interval of plan curvature containing zero (Fig. 8). Even with a 4° slope gradient a portion of the anomaly remains. A low slope threshold may have the unwanted effect of removing some low slope grid cells from along hillsides and within landslides. The use of low slope thresholds should be evaluated based on the conditions in the study area under consideration. All the data points will be used in the analysis presented in this paper.

A comparison of the landslide probability results between the 10-m and 30-m DEM datasets was conducted for a portion of the Parkville and North Kansas City 7½-minute quadrangles. The results of the comparison indicate a shift in the probabilities between the 10-m and the 30-m DEM datasets (Fig. 7b). The 30-m DEM dataset had lower landslide probabilities for the region of plan curvature values shown. However, the relative distribution of landslide probabilities with respect to the intervals of plan curvature remains essentially the same.

3.3. Statistics results

Several landslide maps were used in this study, including 5 published and 1 unpublished landslide-inventory studies in the Kansas portion of the Kansas City metropolitan area (Ohlmacher, 2000a, 2003, 2004a, b, 2005), along with landslide-inventory studies from Marietta, Ohio (Pomeroy, 1984), and Wheeling, West Virginia (Lessing et al., 1976; Davies and Ohlmacher, 1978). Table 1 contains numbers and areal extent of landslides from the landslide-inventory studies. The Marietta, Ohio, study area includes only recent earthflow features. The Wheeling, West Virginia study area was mapped as part of two different landslide studies. Lessing et al. (1976) produced a map of West Virginia portion of the Wheeling quadrangle as part of a project to map landslides in 28 7½-minute quadrangles in the state. Davies and Ohlmacher (1978) mapped the entire Wheeling quadrangle as part of a study mapping landslide features in the Appalachian Plateau from Pennsylvania to northern Alabama. The total area mapped as recent and older landslide is smaller; the individual recent and older landslides are smaller (Table 1); and the area mapped as susceptible to landslides is larger on the landslide map of Wheeling by Lessing et al. (1976) relative to the landslide map by Davies and Ohlmacher

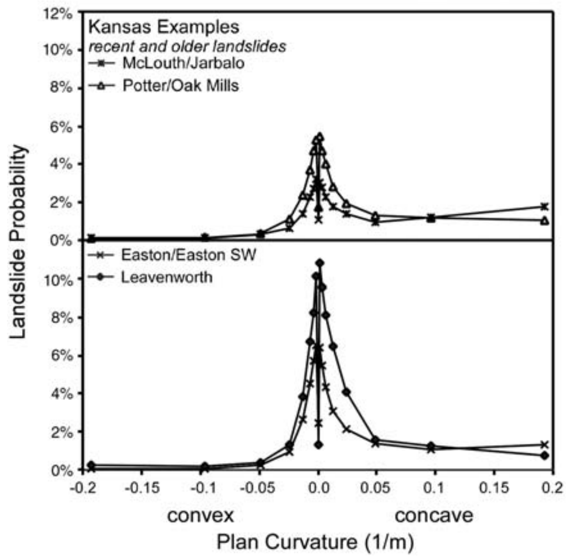

Fig. 9. Plots of landslide probability versus plan curvature for landslide-inventory datasets from northeastern Kansas.

(1978). Although results from both mapping projects are shown, it is beyond the scope of this study to evaluate the quality of the two Wheeling landslide-inventory maps.

Plots of the landslide (earth flow and earth slide) probability with respect to plan curvature are shown in Figs. 7–10. The Kansas results show that hillsides with planar plan curvature have the highest probability of landslides and that areas with concave curvature (hollows) have slightly higher probabilities than areas with convex curvature (Figs. 7–9). Differences in the maximum landslide probability for each individual area may represent the effects of other factors such as changes in material properties and the distribution of slope gradients on landslide susceptibility. An analogous pattern of landslide probability with respect to plan curvature is observed in the recent landslide data for Marietta and Wheeling (Fig. 10a). However, when both the recent and older landslides are analyzed for the Wheeling map by Lessing et al. (1976), the maximum landslide probability is shifted into the region of concave slopes (Fig. 10b). For larger plan curvatures values (more concave), the probability of landslides decreases as in all of the other study areas. The cause of the shift in the maximum probability of landslides in the dataset by Lessing et al. (1976) was not investigated in this study. The landslide-probability results from the landslide-inventory map of Wheeling, West Virginia by Davies and Ohlmacher (1978) (Fig. 10b) are similar to those observed in Kansas and Marietta (Figs. 9 and 10a). In general, two conclusions can be drawn from the statistical analysis: 1) hillsides with planar plan curvature have the highest susceptibility to landslides, and 2) the hillsides with concave curvature are slightly more susceptible to landslides than hillsides with convex curvature.

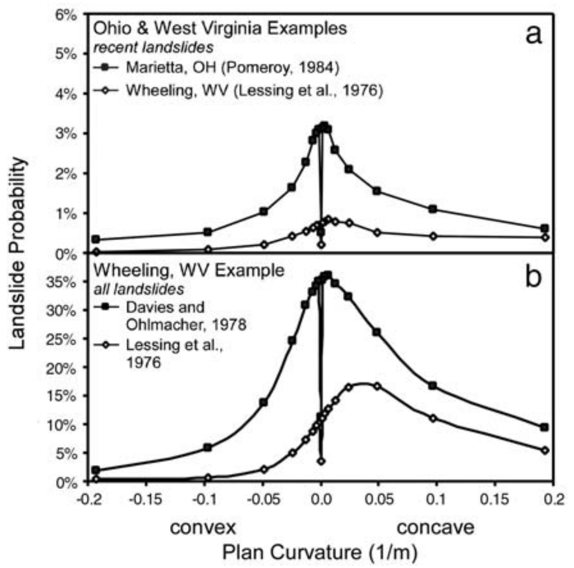

Fig. 10. Plots of landslide probability versus plan curvature for landslide datasets from (a) Marietta, OH and Wheeling, WV using only the recent landslide data and (b) Wheeling, WV using both recent and older landslide data.

4. Discussion

What causes the decrease in susceptibility as the plan curvature becomes more concave? Carson and Kirkby (1972) observed that plan curvature affects the downslope movement of soil in two ways. Where plan curvature is concave, water flow is concentrated into the hollow, which will increase the moisture content of the soil and the amount of time the soil will remain saturated. This, in turn, will increase erosion and decrease stability of the soil. However, soil being transported downslope by various mechanisms including slope wash, creep, and landsliding will also converge in areas with concave plan curvature causing the soil to accumulate in the hollow and stabilizing the slope. Carson and Kirkby noted that the rate of ground-surface lowering is directly related to the downslope sediment flux and inversely related to the curvature (concavity positive). Thus, the rate of ground-surface lowering should be less in hollows.

The convergence of ground water in areas with concave plan curvature is one important factor. Anderson and Burt (1978a,b) installed a series of piezometers in a hollow and measured the distribution of pore-water pressures. Their results show that ground-water flow does converge within an area of concave plan curvature. A combination modeling and field study for an isolated hollow showed that areas of convergent topography (concave plan curvature) tend to remain saturated between storms due to convergence of ground-water flow (Jackson and Cundy, 1992). O'Loughlin (1986) modeled a complex topography with several hollows by examining the upslope catchment area associated with adjacent segments of contours. In hollows, the catchment area is larger than on noses because water flow is converging in the hollow. The results showed that areas with the highest plan curvature saturate first and as the duration of the storm increases, areas with lower plan curvature values begin to saturate. Areas that become saturated first and remain saturated should have higher pore-water pressures and should be more susceptible to landslides. The ground-water model of O'Loughlin was combined with a simple one-dimensional slope stability model in order to determine areas susceptible to debris flows (Montgomery and Dietrich, 1994). The model

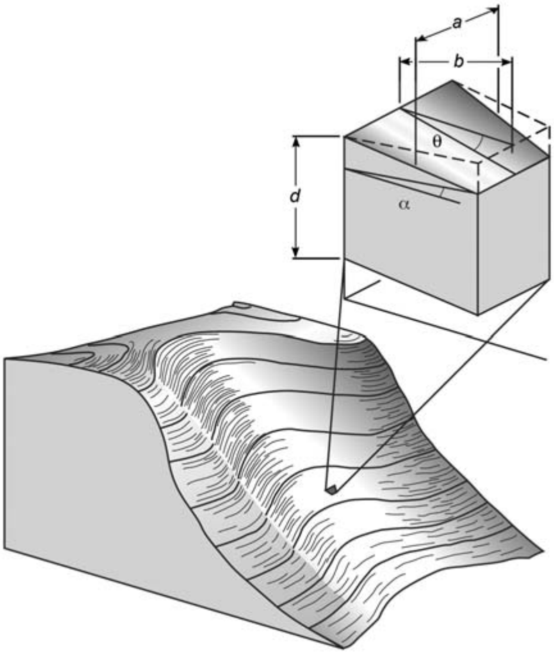

Fig. 11. Schematic drawing of a hollow and the trapezoidal column used as part of the factor of safety analysis.

calculates the minimum steady state rainfall required for a region of the slope to fail. The results indicate that steep channel areas in hollows will fail first. These studies indicate that within a hollow the areas with the highest plan curvature (most concave) should be the most susceptible. As the plan curvature decreases, the susceptibility to landslides should also decrease.

The above discussion only incorporates the effects associated with the convergence of water in the hollow. In a manner analogous to the convergence of water, the material of the landslide is attempting to converge as it moves downslope in a hollow. The convergence of material will lead to lateral forces on the sides of the particles (out-of-the-plane of landslide motion). As adjacent soil grains in the soil mantle are attempting to move in the direction of steepest decent, the grains will converge in areas of concave plan curvature and diverge in areas of convex plan curvature. Additionally, in the study areas the earth flows and earth slides are occurring in fine-grained soils containing clay minerals and cohesion is also adding lateral forces to the analysis.

Lateral forces have a role in slope-stability analyses where end effects are included. Hutchinson (1961) included friction acting on the lateral boundaries of landslides in a list of forces resisting slope failure. Field evidence from studies of mudflows in Kent, England indicates that displacement is confined to a narrow zone along the lateral boundaries and that faults are observed between the mudflow and the undisturbed soil outside the landslide (Hutchinson, 1970). Slope stability analysis that includes the end effects of the landslide indicate that incorporating forces acting on vertical sides of the landslide increases the factor of safety (Baligh and Azzouz, 1975; Hutchinson and Del Prete, 1985; Gens et al., 1988; Hungr et al., 1989). However, the lateral forces acting on vertical surfaces within landslides — for example, columns used in three-dimensional analysis of landslides, appear to be insignificant in slope stability analysis (Hungr, 1987).

The role of lateral forces with respect to relative susceptibility to landsliding is evaluated in this paper by examining the forces acting on a column with a unit surface area (Fig. 11). The column represents a unit slope element and is not an element of a discretized landslide in a slope stability analysis. Slope stability methods that include the end effects exist to determine the factor of safety and potential failure surface for individual landslides (Baligh and Azzouz, 1975; Hungr, 1987; Gens et al., 1988; Chen et al., 2001; Xie et al., 2003). The analysis presented below is developed solely as a measure of the effect of lateral forces associated with plan curvature on landslide susceptibility.

The column has a trapezoidal shape in plan view and is very small compared to the area of the hillside with concave plan curvature. For this analysis, the column is assumed to be at rest. The potential direction of motion of the column is perpendicular to the parallel ends of the trapezoid and from the longer end toward the shorter end (Fig. 12). For the purpose of this analysis, the parallel ends of the trapezoid can be thought of as contours; however, it must be remembered that contours in nature

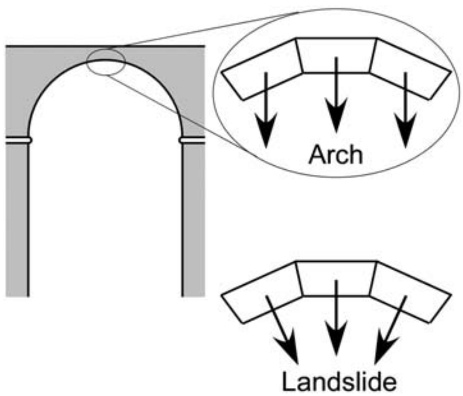

Fig. 12. Comparison of an arch to the motion of the trapezoidal columns of soil within a landslide. For an arch, the blocks are being driven downward by gravity; where as, the plan curvature of the hollow causes the trapezoidal columns to converge.

are rarely parallel. Adjacent to the slanted sides of the trapezoid are additional trapezoidal columns. These columns are rotated relative to the central column, and the direction of movement of the adjacent trapezoidal column relative to the central column leads to convergence of the columns. An equivalent model could be developed using square columns and accounting for the convergence between adjacent columns. The downslope driving force for the column was determined by using the resultant force in the downslope direction parallel to the basal failure surface and is based on the weight of the column. The stress analysis used to determine the lateral forces is somewhat analogous to that for an arch except in the case of the arch, the driving forces for each block are parallel (gravity); where as, the downslope driving forces for the trapezoidal columns converge (Fig. 12). A factor of safety equation for the trapezoidal column was developed using an approach similar to the infinite-slope approach and is:

$$\mathrm{FS} = \frac{(Cab/\cos\theta) + (W-U)\cos\theta\tan\phi + 2(Cdb/\cos\alpha) + 2(W-V)\sin\theta\sin\alpha\tan\phi}{W\sin\theta} \tag{2}$$

where C is the cohesion, a is the average width (perpendicular to motion) of the trapezoid, b is the horizontal length of the trapezoid between the parallel ends, d is the vertical thickness of the column, W is the weight of the column, U is the water force acting along the base, V is the water force acting along the side, θ is the slope gradient, and ϕ is the angle of internal friction. The convergence angle α is the angle between a slanted side of the trapezoid and a side of an equivalent rectangle. The derivation of this equation is presented in the Appendix. An approximate relationship between the convergence angle and the plan curvature κ is given by:

$$\kappa = 2\tan\alpha$$

As was stated earlier, the residual cohesion in soils from landslides in the Appalachian Plateau is very close to zero. Without cohesion, Eq. (2) simplifies to:

$$\mathrm{FS} = \frac{(W-U)\cos\theta\tan\phi + 2(W-V)\sin\theta\sin\alpha\tan\phi}{W\sin\theta} \tag{3}$$

If the convergence angle α is zero (planar plan curvature), Eq. (3) simplifies to the standard factor of safety equation for a cohesionless soil.

Eqs. (2) and (3) indicate that as the convergence angle (plan concavity of a hillside) increases, the lateral resisting forces and the factor of safety increase. The first two terms in the numerator of Eq. (2) and the first term in the numerator of Eq. (3) are the resisting forces along the base of the column. The third and fourth terms in numerator of Eq. (2) and the second term in the numerator of Eq. (3) are the lateral resisting forces on the slanted sides of the trapezoidal column. In a cohesive soil on a hillside with planar plan curvature ($\alpha=0$), a lateral resisting force will still exist as is shown by the third term in Eq. (2). The solid line in Fig. 13 shows the percentage increase in factor of safety with increasing convergence angle for a case of a cohesionless soil with an angle of internal friction ϕ of 27° and with a slope gradient θ of 27°. The water forces U and V were ignored in Fig. 13 because the determination of the water forces will require an understanding of the distribution of hydraulic heads and pore-water pressures along with details of the direction of ground-water migration. The soil density is 20 kN/m^3, and the thickness of the landslide is 2 m. Changes in the density of the landslide material and the thickness of the landslide have no effect on the results for a cohesionless soil. Over the range of convergence angles considered the percentage increase in factor of safety is approximately a straight line, because the increase in factor of safety is controlled by the sine of the convergence angle. For convergence angles greater than those used in Fig. 13, the rate of percentage increase in the factor of safety should slow and the line will no longer be straight.

The effect of cohesion was evaluated by setting a cohesion value equal to 3 kPa, which is the maximum value for the residual cohesion in the Appalachian Plateau. The dashed line in Fig. 13 is for this cohesive soil and indicates that the change in factor of safety is not as great as in the cohesionless soil. However, two issues render the percentage change in factor of safety (Fig. 13) slightly misleading. The magnitude of the factor of safety for planar plan curvature ($\alpha=0$) for the cohesive soil is about 130% higher than an equivalent cohesionless soil because for the cohesive soil, cohesion acts on the sides of the trapezoidal blocks. Also, the magnitude of the change in factor of safety ($\mathrm{FS}_{\alpha=25}-\mathrm{FS}_{\alpha=0}$) is roughly the same for the two soils: 0.43 for the cohesionless soil and 0.55 for the cohesive soil. The slight increase in magnitude for the cohesive soil is because as the convergence angle increases the surface area of the sides of the trapezoidal column increases and the resistance due to cohesion increases along with the frictional resistance. Thus, the addition of cohesion does increase the factor of safety as the convergence angle increases, and the magnitude of the increase in factor of safety is larger for the cohesive soil.

How do these results relate to debris flows? This study lacks data from regions with debris flows. Debris flows can initiate in areas with hollows and in areas with

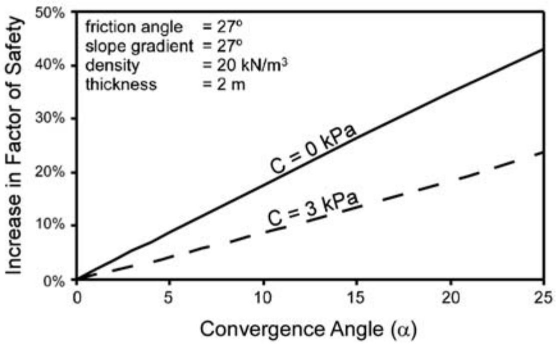

Fig. 13. Plot showing the percentage increase in factor of safety with increasing convergence angle.

planar plan curvature. Some of the 1976 Dorset Mountain debris flows in Vermont (Fig. 1) initiated in regions of the mountainside with planar plan curvature (Fig. 14). The mobilized landslide material moved into a stream channel and continued down the mountain in the stream channel. Insufficient data exists at Dorset Mountain to do a statistical analysis on plan curvature. In the Blue Ridge Province of Virginia where debris flows dominate, Wieczorek et al. (1997) also observed higher landslide susceptibility on hillsides with planar plan curvature. A study of debris slides in Idaho revealed no relationship between plan curvature and landslide susceptibility (Gritzner et al., 2001).

The preceding results from this study with respect to cohesionless soils suggest that susceptibility to debris flows that begin as landslides should be highest in areas of planar plan curvature. However, other factors that affect the initiation of debris flows may influence the lateral resisting forces. Iverson et al. (1997) indicate that mobilization of debris flows involve widespread Coulomb failure, liquefaction of the material, and conversion of translational energy to vibrational energy. They further state that the factors leading to liquefaction include an upward component of ground-water flow and a densification of the material with a resulting increase in pore-water pressure. The process of densification and increase of pore-water pressure might lead to a reduction in the lateral resisting stresses by reducing the effective stress along the sides of the trapezoidal column. This reduction of resisting forces on the sides will decrease the factor of safety when compared to landslides without liquefaction. Thus, the relationship between plan curvature and debris flows may be different from the relationship for earth flows and earth slides.

This analysis indicates that the relationship between plan curvature and landslide type is complex. A valid landslide-susceptibility or landslide-hazard analysis that includes plan curvature requires knowledge of the local relationship between plan curvature and landslides. This, in turn, requires quality landslide-inventory data and quality digital elevation data. From the quality data, the local relationship between plan curvature and landslide susceptibility can be developed. Knowledge of the relationship between plan curvature and landslides is very important in statistical methods like multiple logistic regression where plan curvature can be treated as a continuous variable. It is not as critical in statistical methods where plan curvature is treated as a categorical variable and subdivided into intervals. For the statistical analysis presented above which categorized the plan-curvature values, the optimum results were obtained by using unequal ranges of plan-curvature values.

Landsliding, slope wash, gullying, subsidence, tectonics, creep, and other processes shape the local landform geometry and play a role in the relationship between plan curvature and landslide susceptibility. The region surrounding the Ohio River in Ohio and West Virginia is dominated by a landform consisting of

Fig. 14. Debris flow on Dorset Mountain in Vermont. This 1976 debris flow started in a region of relatively planar plan curvature.

alternating hollows and noses. The same is true for northeastern Kansas, especially in the vicinity of the City of Leavenworth. Some hollows are very broad (Fig. 5d). Contours in these areas have a complex quasi-sinusoidal shape (Fig. 4). Earth flows and earth slides are the dominant slope process in these regions. Areas with planar plan curvature are the most susceptible to landslides. Regions of the hillside that have developed slight concavity because of landslide events will be more susceptible than adjacent regions with slight convexity. The hollows will increase in size, and a hollow and nose terrain will develop. Thus, areas with planar plan curvature should not remain planar. Understandably, slope wash, gullying, and creep are active in both study areas. The contribution of karst in the study areas is very minor because the thicknesses of the limestone and other karst-prone units are too thin relative to the local relief to create sinkhole features of this magnitude. When compared to the plate boundaries along the west coast of North America, the contribution of tectonics in this region is relatively minor. The hills in these study areas are most likely being shaped by landslides, slope wash, and creep, and this landform of hollows and noses may be the result. Thus, a landslide hollow may represent an end-member landform in terrain dominated by earth flows and earth slides.

5. Conclusion

The statistical analysis presented above indicates that hillsides with planar plan curvature are the most susceptible to earth flows and earth slides. Hillsides with concave plan curvature (hollows) are slightly more susceptible to landslides than hillsides with convex plan curvature (noses). Although conventional wisdom suggests that the convergence of ground and surface water in hollows should increase landslide susceptibility, the resisting forces between soil particles that results from convergence of material increases the stability of hollow. The type of landslide along with other geomorphic processes may play a role in the nature of the local relationship between plan curvature and landslide susceptibility. The local relationship is important to the treatment of plan curvature in statistical landslide susceptibility and hazard analysis. A landform composed of hollows and noses is a relatively stable landform in regions dominated by earth flows and earth slides.

Acknowledgments

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Appendix A

The following section presents the derivation of the equations used in the analysis of plan curvature and landslides. The analysis is based on the infinite slope model for slope stability. A very small block of material is isolated for analysis from the entire slope. For this analysis the block is shaped like a parallelogram in the cross section (side view in Fig. A1) and the top of the block is a trapezoid (map view in Fig. A1). A static's approach is used to create the equation for the factor of safety. The depth of the block is d and extends from the ground surface to the slide plane. The slide plane is parallel to the ground surface. The average width of the block normal to the downslope direction is a (map view in Fig. A1). The horizontal thickness of the block in the downslope direction is b (side view in Fig. A1), and b is very small as compared to the length of the slope in the

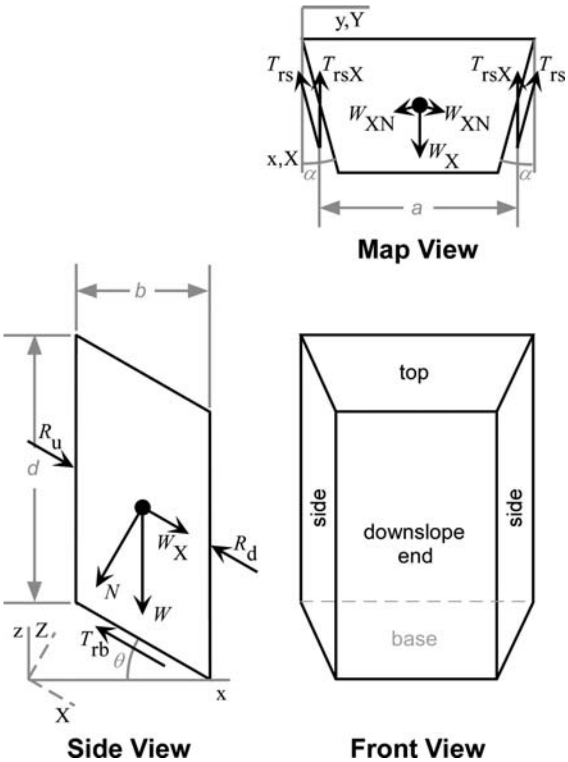

Fig. A1. Sketch of trapezoidal column showing stresses.

downslope direction. The weight W of the block is given by:

$$W = \gamma_s abd$$

where γ_s is the unit weight of the soil. In order to simplify the mathematics the coordinate system will be rotated so that the X and Y axes will be parallel to the slide plane and the Z axis will be normal to the slide plane (Fig. A1). The dip of the slide plane is given by the angle θ. The weight is then resolved into a force normal to the X–Y plane N and a force parallel to the X axis W_X. These force components are given by the following equations:

$$N = W\cos\theta$$

and

$$W_X = W\sin\theta.$$

The material upslope of the block applies a force on the upslope end of the block R_u and a force R_d in the opposite direction is applied to the downslope end of the block by the material downslope of the block. Cohesion and friction along the base and sides create tractions parallel to these surfaces, T_{rb} and T_{rs} respectively. The tractions along the sides are in the X–Y plane but are not in the direction of the X axis (map view in Fig. A1). The convergence angle α is defined as the angle between the X-axis direction and the side of the block. Thus, the traction on the side of the in the X-axis direction T_{rsX} is given by:

$$T_{rsX} = T_{rs}\cos\alpha.$$

The weight of the block as the block attempts to move down the slide plane applies a normal force to the sides of the block W_{XN} (Fig. A1) which is given by:

$$W_{XN} = W_X\sin\alpha = W\sin\theta\sin\alpha.$$

Now, summing the forces and tractions in the X direction yields:

$$\sum F_X = W_X - T_{rb} - 2T_{rsX} + R_u - R_d. \quad (1)$$

The constant 2 in front of the side traction variables accounts for the two sides of the block. Eq. (1) cannot be simplified by taking moments about any point within the block. In order to simplify Eq. (1), the standard assumption of the infinite slope model will be applied. This model assumes that the thickness of the block b is much smaller than the length or that the value of b approaches 0. Here, the magnitudes of the forces on the upslope and downslope ends are equal or:

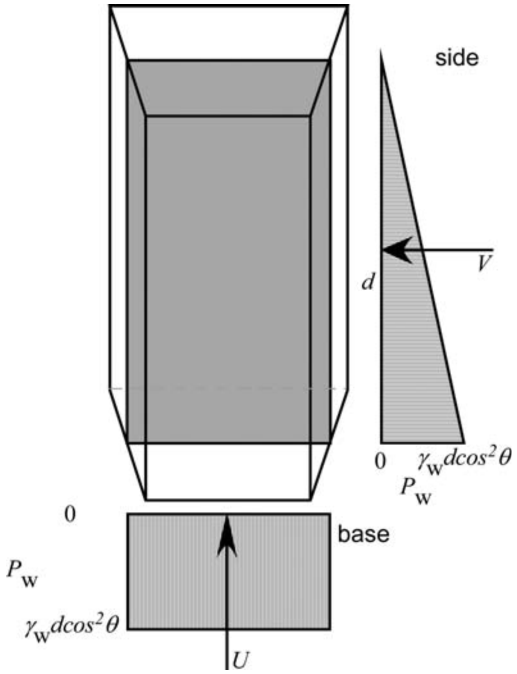

Fig. A2. Water forces acting on the trapezoidal column.

$$R_u \cong R_d.$$

This assumption will apply to blocks with converging sides as long as the convergence angle α is small. If the convergence is greater that 45°, then the assumption may not be valid because the area of the downslope end will be considerably smaller than the upslope end. Applying the assumption to Eq. (1) and assuming the block is at equilibrium yields:

$$\sum F_X = W_X - T_{rb} - 2T_{rsX} = 0 \quad (2)$$

Moving the tractions to the right side of Eq. (2) and dividing both sides by W_X yields the factor of safety equation at equilibrium:

$$\mathrm{FS} = 1 = \frac{T_{rb} + 2T_{rsX}}{W_X}. \quad (3)$$

Using the Mohr–Coulomb failure criteria and removing the equilibrium restriction yields the factor of safety equation:

$$\mathrm{FS} = \frac{(C_{ab}/\cos\theta) + N\tan\phi + (C_{db}/\cos\theta) + 2W_{XN}\tan\phi}{W_X}$$

or

$$FS = \frac{(Cab/\cos\theta) + W\cos\theta\tan\phi + 2(Cdb/\cos\alpha) + 2W\sin\theta\sin\alpha\tan\phi}{W\sin\theta} \tag{4a}$$

Since cohesion C has units of stress, the cohesion value is multiplied by the area of the base or side as appropriate in the above equations. For cohesionless materials, Eq. (4a) simplifies to:

$$FS = \frac{W\cos\theta\tan\phi + 2W\sin\theta\sin\alpha\tan\phi}{W\sin\theta} \tag{4b}$$

The above equations do not include the effects of pore-water pressure on the strength of the slip surfaces. For this analysis of effective normal stress, the block will be assumed to be saturated with slope parallel ground-water flow. Iverson (1990) discusses permissible ground-water flow fields for infinite slopes. Fig. A2 depicts pore-water pressures along a plane through the center of the center of the block. The pore-water pressure is constant everywhere along the base of the block and is equal to the unit weight of water γ_w multiplied by $d\cos^2\theta$. On the sides of the block, the pore-water pressure varies from the magnitude at the base to zero at the top. Resultant water forces are calculated for each of the slip surfaces where U is the water force acting normal to the base and V is the water force acting along each side. The water forces are given by:

$$U = \gamma_w d(\cos^2\theta)a(b/\cos\theta) = \gamma_w dab\cos\theta$$

and

$$V = 1/2\gamma_w d(\cos^2\theta)b(a/\cos\alpha).$$

The factor of safety Eqs. (4a) and (4b) become:

$$FS = \frac{(Cab/\cos\theta) + (W-U)\cos\theta\tan\phi + 2(Cdb/\cos\alpha) + 2(W-V)\sin\theta\sin\alpha\tan\phi}{W\sin\theta} \tag{5a}$$

and

$$FS = \frac{(W-U)\cos\theta\tan\phi + 2(W-V)\sin\theta\sin\alpha\tan\phi}{W\sin\theta} \tag{5b}$$

Eqs. (4b) and (5b) simplify to the standard forms for the infinite slope method if the convergence angle α is set to zero. In a normal infinite slope model it is assumed that the landslide is very wide and the material beside the blocks is also moving. Thus, the cohesion terms associated with the sides is zero in Eqs. (4a) and (5a). In the analysis presented in this paper, the block is assumed to be at rest and in order for it to move, it needs to move material laterally out of its path downslope.

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